# Inverse Problems with Diffusion vs. Flow Matching Posterior Sampling

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## Problem Statement

Recovering clean images from degraded observations is a fundamental inverse problem in computer vision. In this work, we consider the standard linear degradation model:

$$y = Ax + \epsilon$$

where $x$ is the unknown clean image, $A$ represents a blur operator (e.g., convolution with a kernel), and $\epsilon$ denotes additive noise. Recent approaches such as *Diffusion Posterior Sampling* (DPS) leverage pretrained diffusion models (e.g., DDPMs) as powerful generative priors, combined with gradient-based measurement guidance to solve this inverse problem. While DPS achieves strong reconstruction quality, it typically requires hundreds of neural function evaluations (NFEs), leading to high computational cost at inference time. *Flow Matching* (FM) has recently emerged as an efficient alternative generative modeling framework that can achieve comparable generative performance using significantly fewer ODE integration steps. This project rigorously investigates whether FM-based posterior sampling can match or surpass DPS in reconstruction quality while providing meaningful gains in inference efficiency. In addition to reconstruction accuracy and speed, we examine a critical robustness challenge: model misspecification. In practical scenarios, the assumed degradation operator may differ from the true one (e.g., assuming Gaussian blur when the actual degradation is motion blur), which can lead to significant performance degradation. We therefore evaluate both methods under such out-of-distribution conditions to assess their robustness and reliability in realistic settings.

## Proposed Approach

We plan to conduct a controlled comparison of two posterior sampling paradigms for solving the inverse problem, ensuring identical experimental conditions and using available pretrained checkpoints.

**Diffusion Posterior Sampling (DPS):** A pretrained diffusion model (DDPM) on FFHQ-256 is combined with gradient-based measurement guidance, where the likelihood term is injected at each reverse diffusion step. While effective, this approach requires a large number of neural function evaluations (approximately 100–1000 NFEs) due to stochastic reverse SDE sampling.

**Flow Matching (FM) with FlowDPS guidance:** A pretrained flow-matching model is used with ODE-based velocity correction for posterior sampling. This replaces the stochastic reverse process with a deterministic ODE formulation, enabling substantially more efficient sampling (approximately 25–100 NFEs) while maintaining comparable generative performance. Both methods operate on the same degraded inputs generated from a shared forward model. The sampling procedure is the only variable, while the architecture, dataset, and degradation settings are held constant, ensuring a fair and controlled evaluation.

## Experimental Design

| Parameter | Values | Conditions | Runs |
|---|---|---|---|
| Gaussian blur $\sigma$ | 1.5, 3.0, 5.0 | 3 levels | |
| Noise $\sigma$ | 0.00, 0.05 | 2 levels | |
| Sampling steps | 25, 50, 100 | 3 configs | $36 \times 2$ |

Table 1: Experimental configurations.

### Flow Diagram

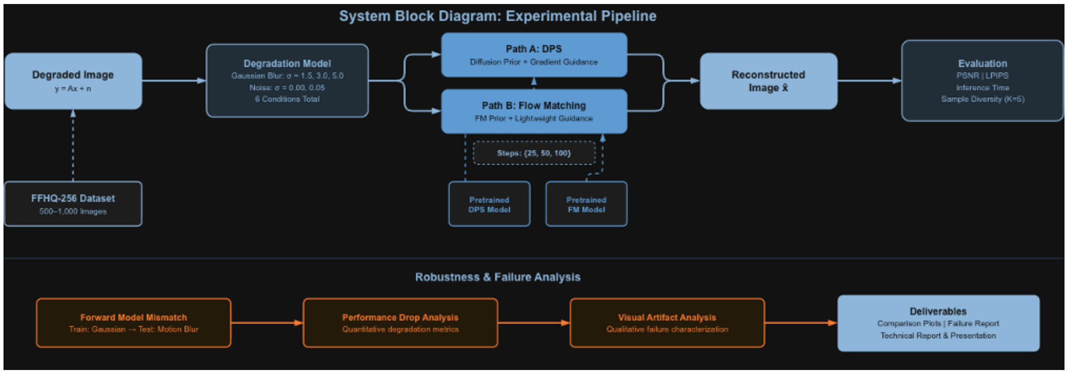


Figure 1: Flow diagram of the image reconstruction pipeline. The degraded observation $y$ is generated from the clean image $x$ using the forward model. Two posterior sampling approaches, DPS and Flow Matching (FM), are then applied to recover an estimate of the clean image.